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Roll No

EX-8003 (3) (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Special Machine
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Discuss the torque -pulse rate characteristics of stepper motor. Enumerate the features that the drive circuit for a stepper motor should possess for optimum torque output.
2. Give the principle of operation of switched reluctance motor. How the torque is produced explain in brief with the help of expression? Also explain shaft position sensing.
3. Explain the construction and principle of operation of brush less DC motor .What do you mean by winding pattern series and parallel?
4. a) Explain how the speed of brushless DC motor is controlled?
b) What are the different applications of permanent magnet DC motor?
5. Write a short note on the following.
 - a) Design parameter of DC brushed motor
 - b) Permanent magnet material
6. Explain the PM synchronous motor construction and operation along with speed control and give its applications.
7. a) Write the comparison between variable reluctance and permanent magnet stepper motor.
b) What are the important features of stepper motor?
8. Explain the following (any two)
 - a) Power electronic converter of stepper motor.
 - b) Sensor less control of permanent magnet synchronous motor.
 - c) Advantage and disadvantage of solid rotors.
